**Exercise 11.10**

(1) (i) $\frac{1}{4}\log\left|\frac{2+x}{2-x}\right|+c$ (ii) $\frac{1}{20}\log\left|\frac{5+2x}{5-2x}\right|+c$ (iii) $\frac{1}{12}\log\left|\frac{3x-2}{3x+2}\right|+c$

(2) (i) $\frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}-3+x}{\sqrt{2}+3-x}\right|+c$ (ii) $\frac{1}{10}\log\left|\frac{x-4}{x+6}\right|+c$ (iii) $\log\left|x+2+\sqrt{x^2+4x+2}\right|+c$

(3) (i) $\log\left|x+2+\sqrt{(x+2)^2-1}\right|+c$ (ii) $\log\left|x-2+\sqrt{x^2-4x+5}\right|+c$ (iii) $\sin^{-1}\left(\frac{x-4}{5}\right)+c$

**Exercise 11.11**

(1) (i) $\log\ |x^2+4x-12|-\frac{7}{8}\log\left|\frac{x-2}{x+6}\right|+c$ (ii) $\frac{5}{2}\log|x^2+2x+2|-7\tan^{-1}(x+1)+c$

(iii) $\frac{3}{4}\log|2x^2-2x+3|+\frac{\sqrt{5}}{2}\tan^{-1}\left(\frac{2x-1}{\sqrt{5}}\right)+c$

(2) (i) $5\sin^{-1}\frac{x-2}{\sqrt{13}}-2\sqrt{9+4x-x^2}+c$ (ii) $\sqrt{x^2-1}+2\log\left|x+\sqrt{x^2-1}\right|+c$

(iii) $2\sqrt{x^2+4x+1}-\log\left|x+2+\sqrt{x^2+4x+1}\right|+c$

**Exercise 11.12**

(1) (i) $\frac{x+1}{2}\sqrt{x^2+2x+10}+\frac{9}{2}\log\left|x+1+\sqrt{x^2+2x+10}\right|+c$

(ii) $\frac{x-1}{2}\sqrt{x^2-2x-3}-2\log\left|x-1+\sqrt{x^2-2x-3}\right|+c$

(iii) $\frac{x-5}{2}\sqrt{10x-x^2-24}+\frac{1}{2}\sin^{-1}(x-5)+c$

(2) (i) $\frac{1}{4}\left[(2x+5)\sqrt{9-(2x+5)^2}+9\sin^{-1}\left(\frac{2x+5}{3}\right)\right]+c$

(ii) $\frac{1}{4}\left[(2x+1)\sqrt{81+(2x+1)^2}+81\log\left|2x+1+\sqrt{81+(2x+1)^2}\right|\right]+c$

(iii) $\frac{x+1}{2}\sqrt{(x+1)^2-4}+2\log\left|x+1+\sqrt{(x+1)^2-4}\right|+c$

**Exercise 11.13**

| (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) | (13) | (14) | (15) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 4 | 1 | 3 | 1 | 3 | 2 | 4 | 3 | 4 | 2 | 4 | 4 | 1 |
| (16) | (17) | (18) | (19) | (20) | (21) | (22) | (23) | (24) | (25) | | | | | |
| 1 | 3 | 2 | 4 | 1 | 3 | 4 | 3 | 1 | 4 | | | | | |



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